India Hydrological Outlook

Based on daily observations till 28th February 2026

Issue date: 01.03.2026

SUMMARY: In February, India faced very low rainfall across north, northeast, northwest, and east regions, while the south showed mixed conditions leaning toward higher-than-normal amounts; central areas recorded lower-than-normal precipitation. Temperatures were warmer-than-normal in north, northeast, northwest, and west regions, with cooler-than-normal conditions in the south, and near-normal readings elsewhere. Relative wetness varied, with the west experiencing wetter-than-normal soil moisture and mixed conditions leaning toward drier-than-normal levels in the northeast and north. Total runoff exhibited a high deficit in the north and northeast, near-normal levels elsewhere. Evapotranspiration showed reduced conditions in the north and northeast, while most other regions remained near-normal. Streamflow in rivers and reaches was very high in the west and south, very low in the northeast, and mixed leaning toward lower-than-normal in the north. In March, higher-than-normal rainfall is forecast for the northeast, northwest, and north, near-normal for the central, east, and south, and very high for the west.

Rainfall and Temperature:

In February, the majority of northern, central, and eastern India recorded very low rainfall, while the south experienced mixed conditions leaning toward higher amounts. By March, higher-than-normal rainfall is predicted in the north, northeast, and northwest, whereas the central, east, and south are expected to receive near-normal to slightly elevated precipitation. Temperatures were warmer than normal across northern, northeastern, and western India in February, though the south was cooler.

Soil moisture, Total runoff, and Evapotranspiration:

In February, soil moisture was relatively higher than normal in west India, while near-normal conditions prevailed across the rest of the country. In March, soil moisture is expected to remain near-normal across most regions, with drier-than-normal conditions leaning toward the northeast. A high runoff deficit was observed in northern India in February, while near-normal runoff is anticipated in March across the majority of the country. Evapotranspiration in February was reduced in east-central India, but conditions are mixed elsewhere.

River flows:

In February, very high streamflow was observed across the south and west, while the northeast experienced very low flow and the north saw mixed conditions leaning toward lower-than-normal levels. March forecasts indicate a sharp reversal, with very high flow predicted in the northeast and northwest, while the central region is expected to shift toward lower-than-normal conditions, contrasting with the near-normal trends seen in the east and central-east areas.

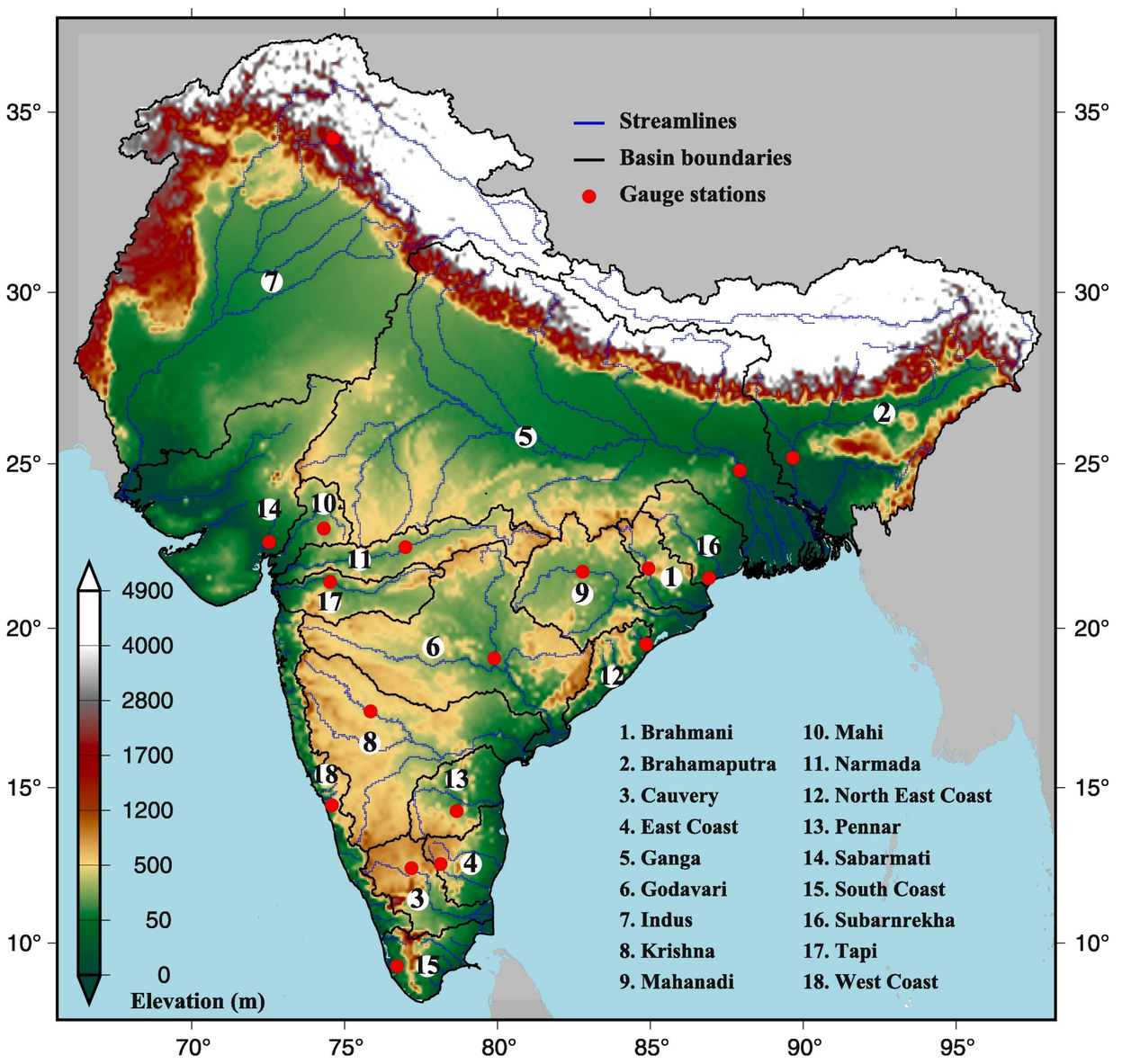


Figure. Indian sub-continental river basins. Shaded background shows elevation (m).

India Hydrological Outlook provides a comprehensive monthly summary of key meteorological and hydrological variables for current conditions alongside a four-month retrospective and one month forecast. For more information, please visit the website: www.indiahydrolook.in

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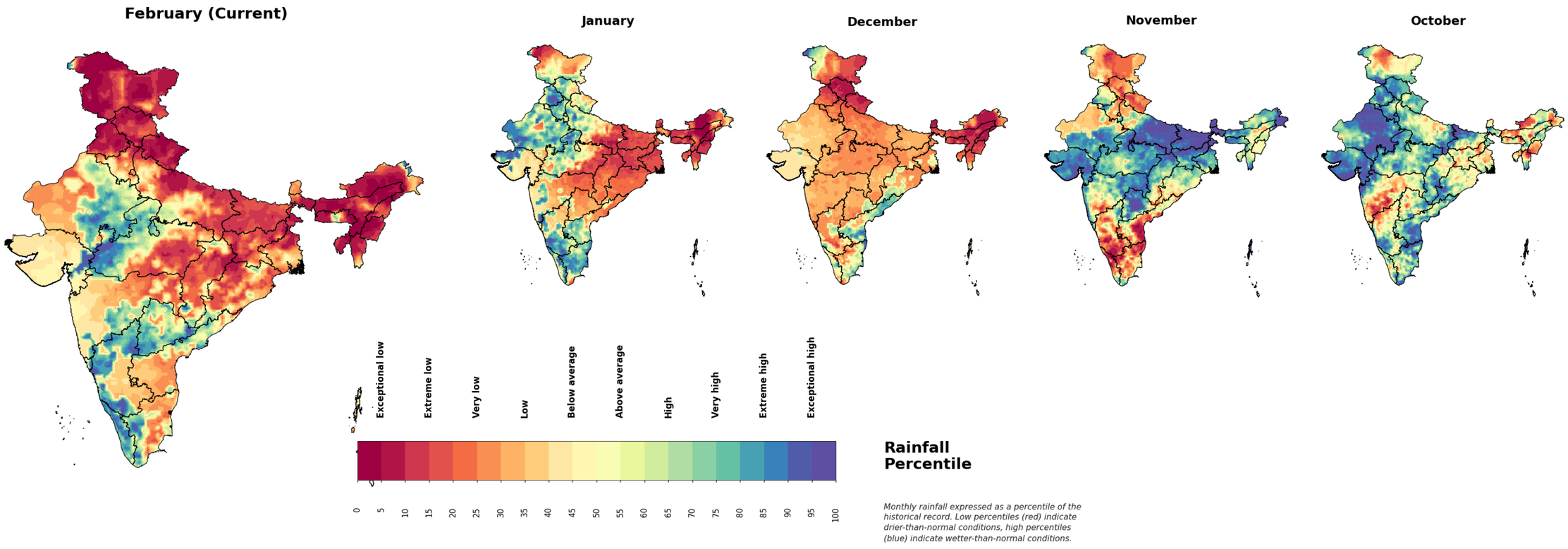
Observed and Forecast Rainfall Conditions

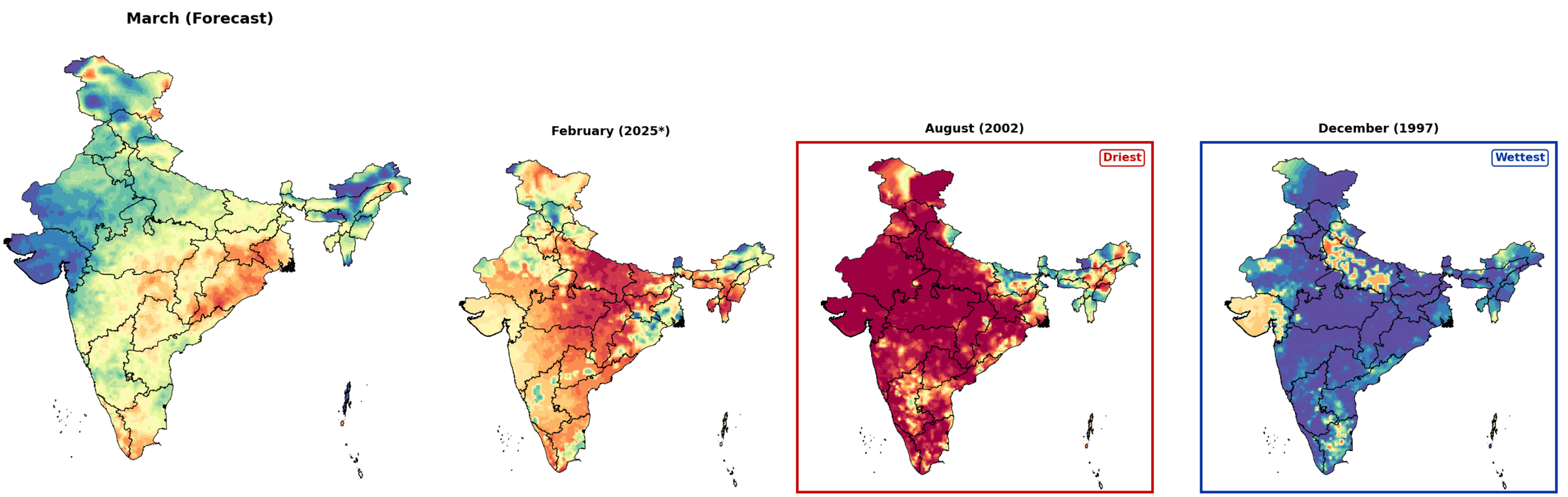
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These maps are prepared from daily gridded rainfall observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as total monthly rainfall in percentile for period 1955 to present. These rainfall percentile maps highlight the regions with unusual dry, wet, or near-average rainfall for that time of year. High rainfall percentiles in forecast month, along with high relative wetness (maps available on page 4) in current month, may indicate a potential region for unusual wet conditions. Areas with low precipitation percentiles in forecast month, particularly those with minimal or no rainfall along with high relative dryness, may be at risk of drought conditions.

Summary: Relatively low rainfall characterized February across India, with very low conditions in the north, northeast, northwest, and east. In March, higher-than-normal rainfall is predicted for the northeast and northwest, while near-normal conditions are forecast for central, east, and southern parts.





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Observed and Forecast Surface Air Temperature

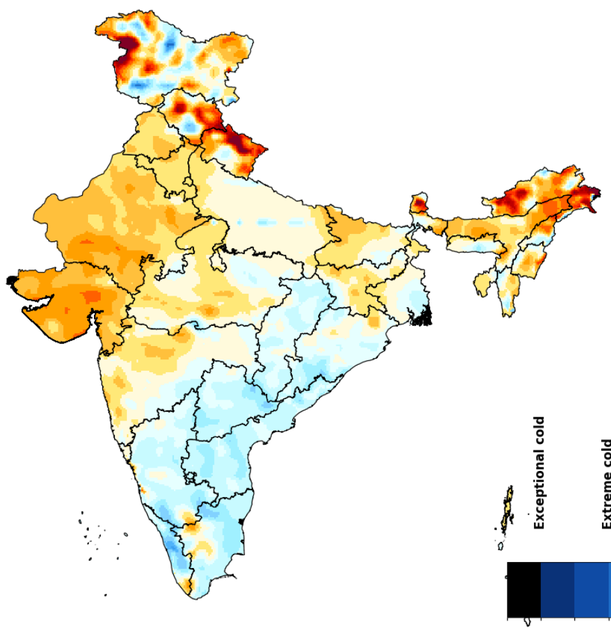
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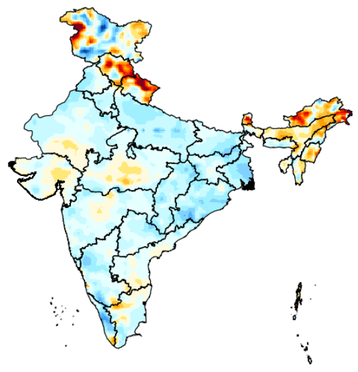
These maps are prepared from daily gridded temperature observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a monthly-averaged temperature anomaly from the historical mean (1955 to present). These temperature anomaly maps highlight regions with higher and lower temperatures than usual. Low-temperature anomalies may lead to a decrease in crop yield or delayed crop growth. Areas with higher temperature anomalies paired with relatively low rainfall and low soil moisture (maps available on page 4) may be at risk of drought conditions with increased potential evapotranspiration.

Summary: In February, warmer-than-normal temperatures persisted across India except the south and central-east regions, while near-normal conditions prevailed in central areas. In March, much warmer-than-normal temperatures are forecast across India, except the north and northeast where significantly cooler-than-normal conditions are predicted.

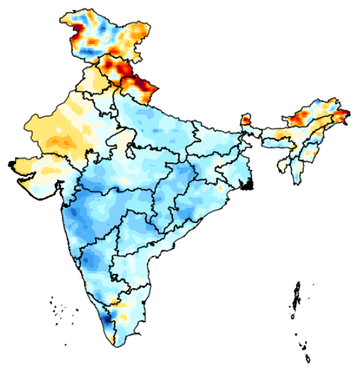
February (Current)



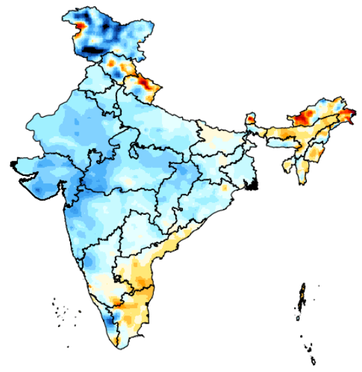
January



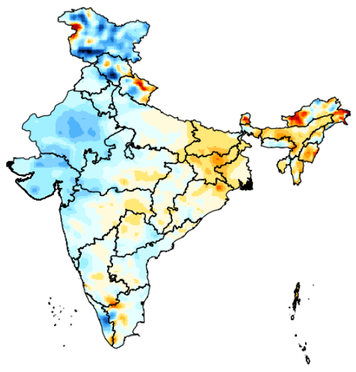
December

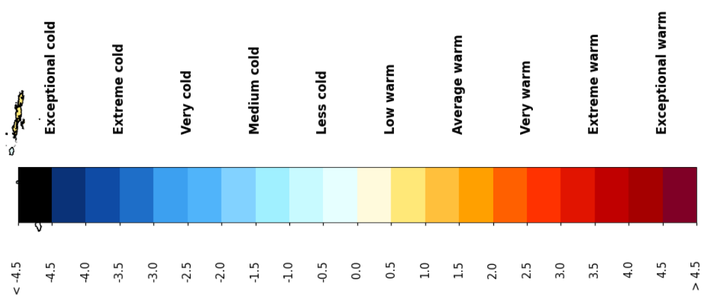


November



October

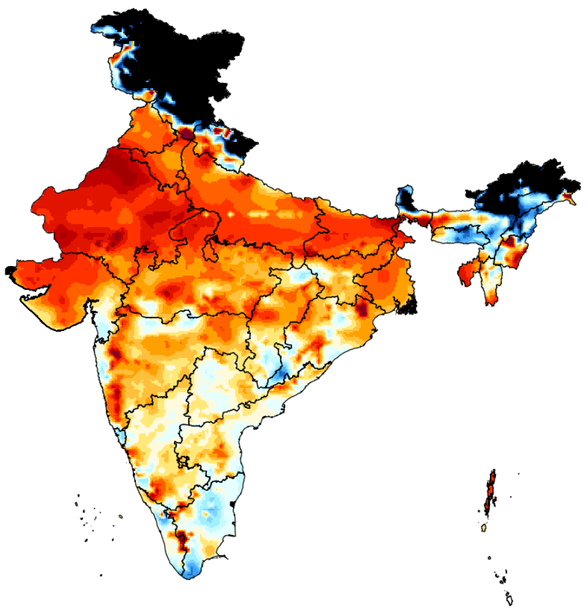




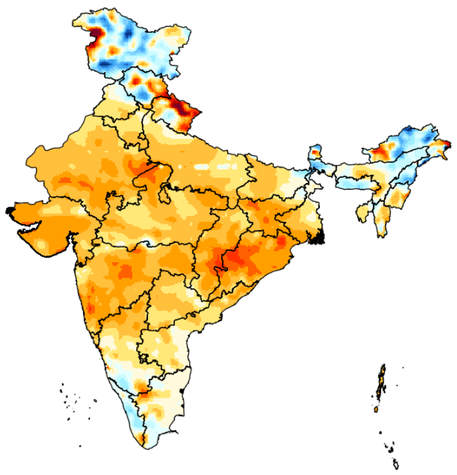
Temperature Anomaly (°C)

Surface air temperature anomaly compared to the long-term mean for the same month. Negative values (blue) indicate cooler-than-average conditions; positive values (red) indicate warmer-than-average conditions.

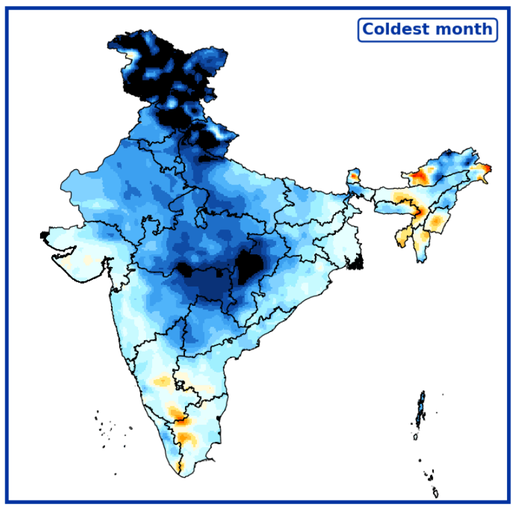
March (Forecast)



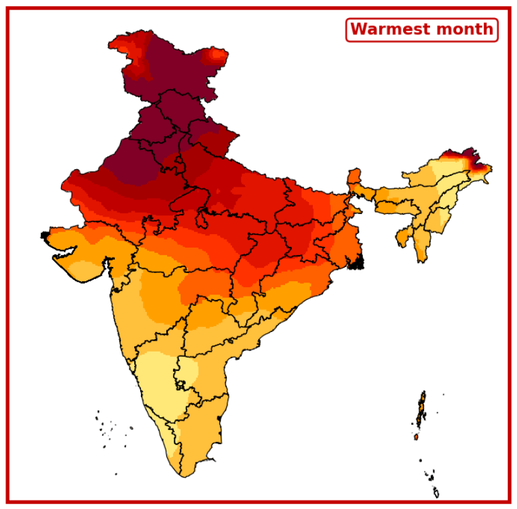
February (2025*)



May (2023)



April (2010)



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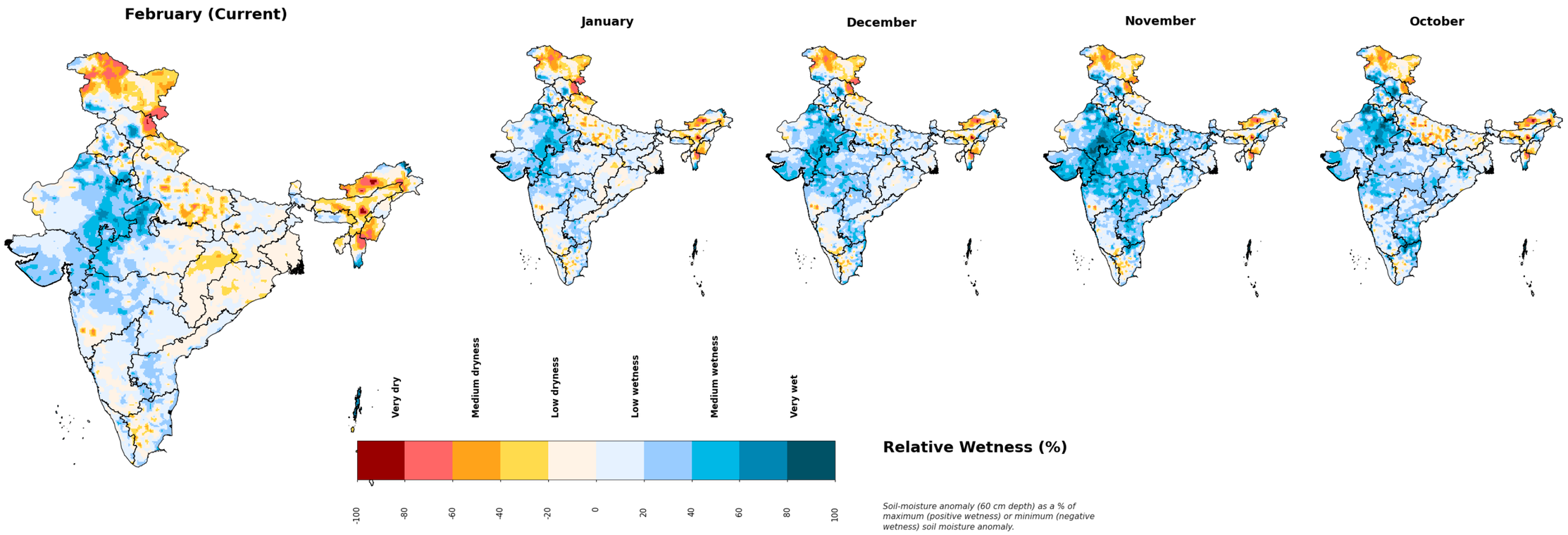
Observed and Forecast Relative Wetness

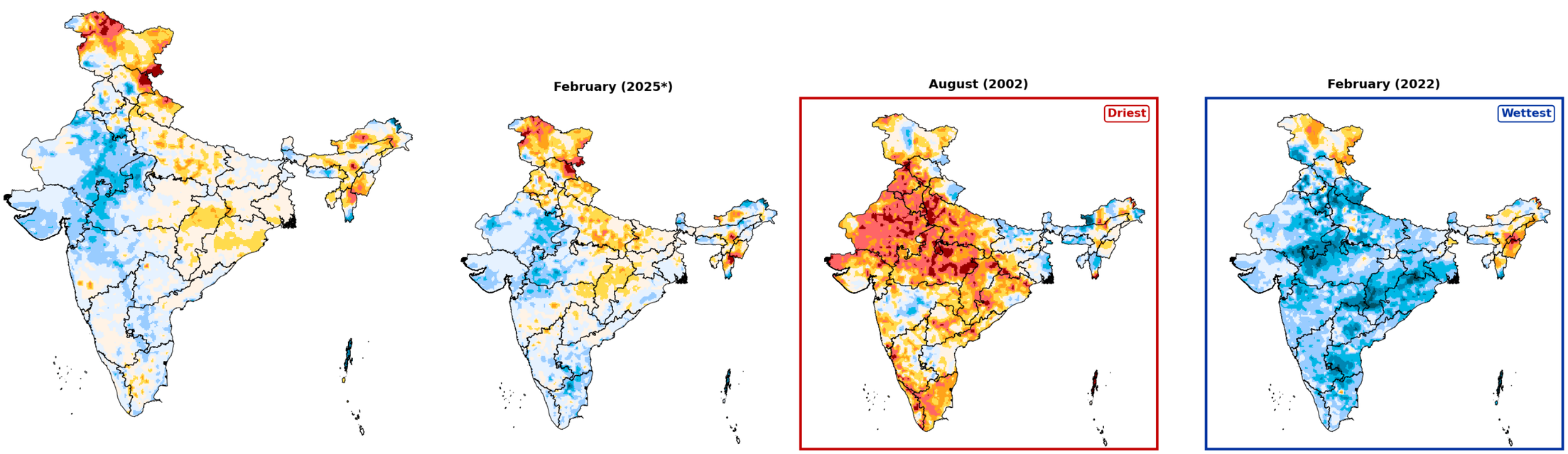
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These maps are prepared from daily simulated soil moisture (60 cm depth) using gridded meteorological forcing from observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a monthly-averaged soil moisture anomaly from the historical mean (1955 to present). The soil moisture anomalies are presented as relative to historical extremes to highlight the regions with unusual wet or dry conditions. High relative wetness (wetter) in current month paired with a high rainfall percentile in forecast month may lead to above-normal flows and unusual wetter conditions in the coming days/weeks. Areas with low relative wetness (drier) in current month and no or minimal predicted rainfall in forecast month may be at risk of lower than normal flow and moisture availability.

Summary: Soil moisture remains near normal across most of India in February, while the west stays wetter than usual. In March, conditions persist near normal across the majority of the country, though the north and northeast lean toward drier-than-normal soil.





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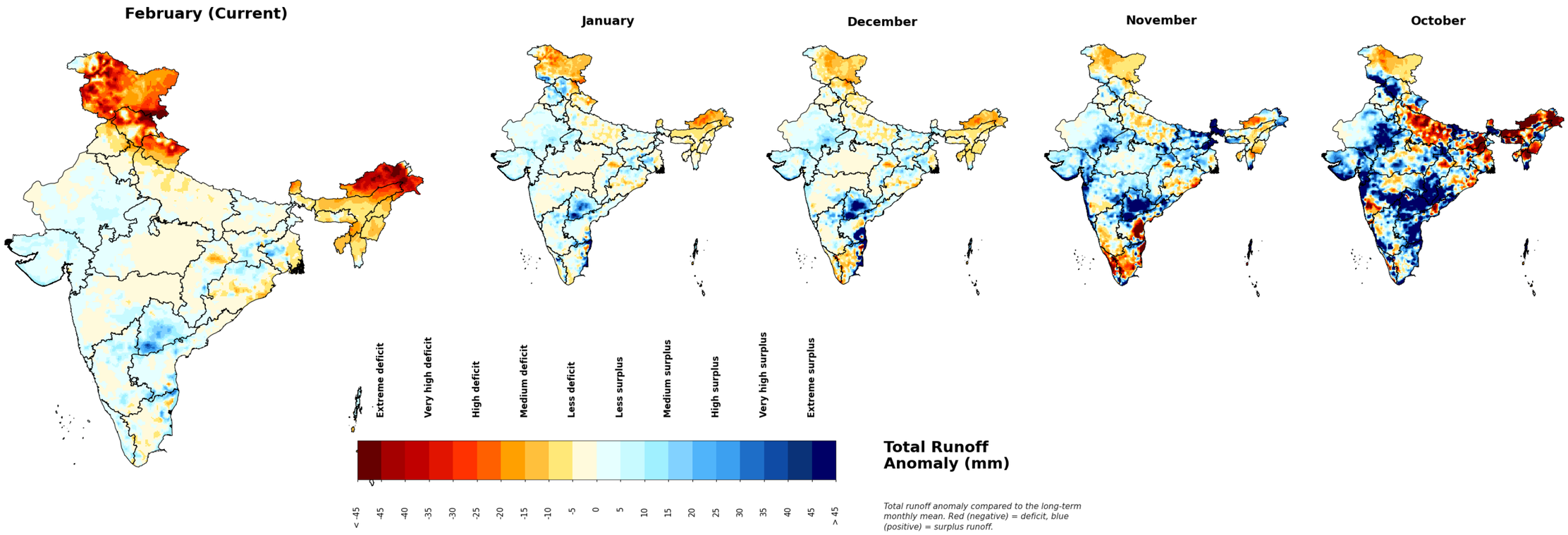
Observed and Forecast Total Runoff

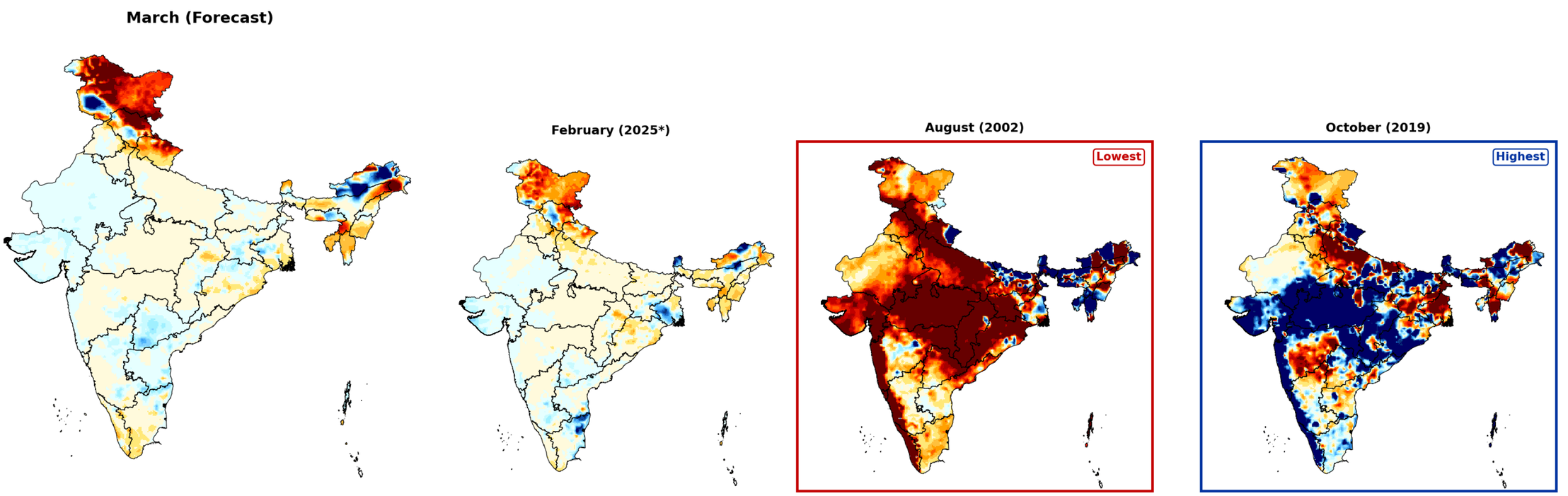
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These maps are prepared from daily simulated total runoff using gridded meteorological forcing from observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a total runoff monthly anomaly from the historic mean (1955 to present). These total runoff anomalies highlight the regions with relative surplus or deficit in total runoff than typical for that time of year. A surplus in total runoff in forecast month paired with relatively higher soil moisture in current month may lead to above-normal streamflow in the coming days/weeks. Areas with a high deficit in total runoff in forecast month, particularly those with high dryness in current month, may be at risk of relatively low streamflow in the coming days/weeks.

Summary: In February, near-normal total runoff persisted across India except north and northeast, where a high runoff deficit was observed.





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Observed and Forecast Evapotranspiration

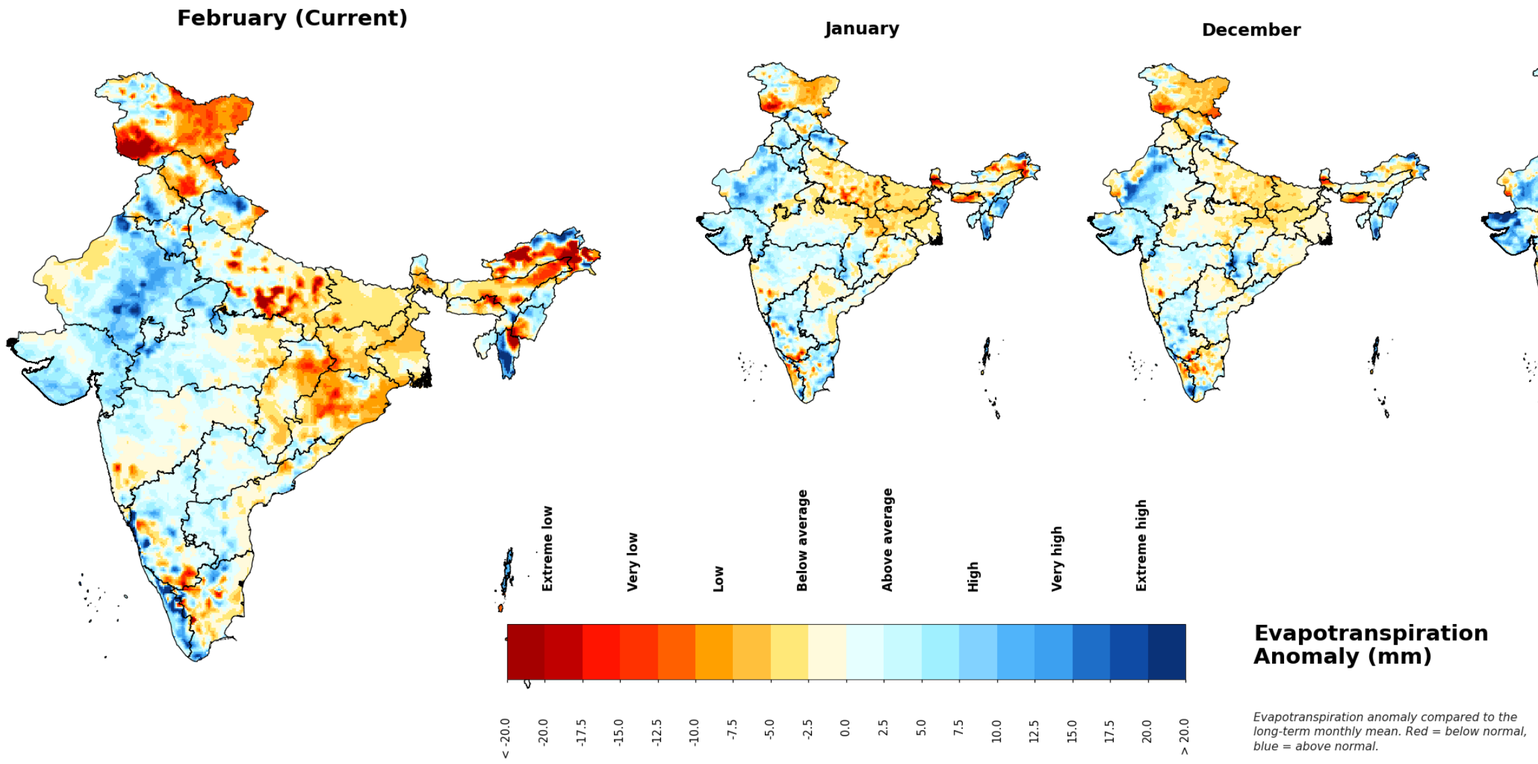
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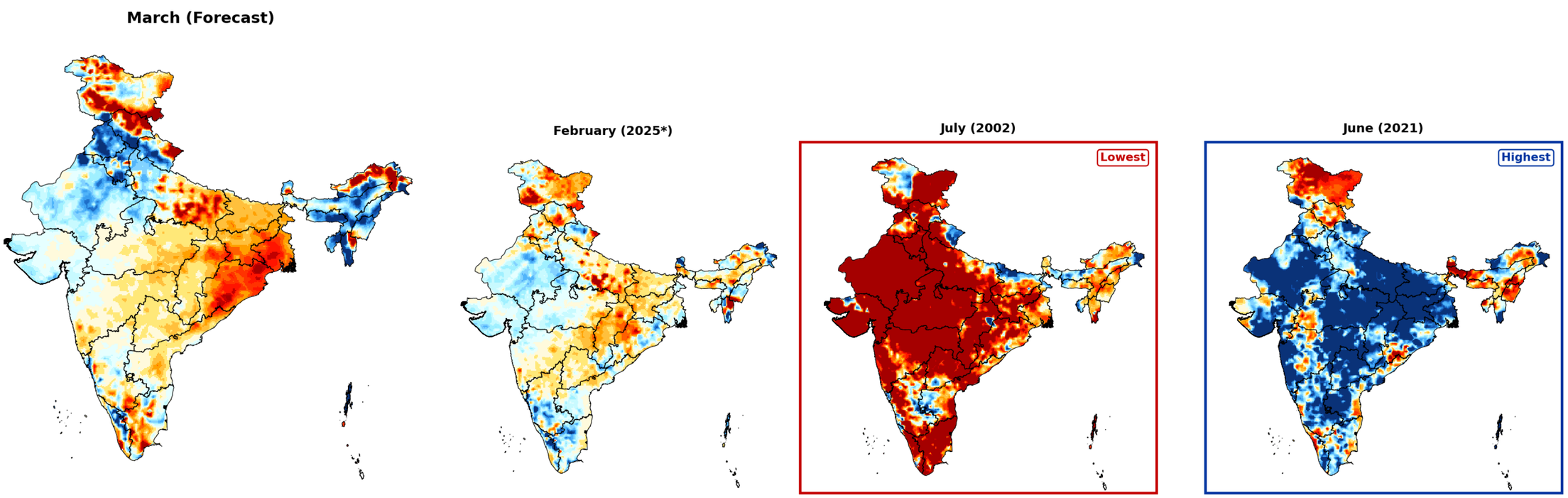
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These maps are prepared from daily simulated evapotranspiration (ET) using gridded meteorological forcing from observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a total monthly ET anomaly from the historic (1955 to present). These ET anomalies highlight the regions with less or more water lost than usual from land and plants, helping to pinpoint areas at risk of extreme weather impacts, especially crop water stress. Low ET anomalies indicate increased crop water stress and relatively less moisture availability. High ET anomalies may lead to rapid depletion of soil moisture (flash drought) or may worsen the pre-existed droughts.

Summary: In February, mixed conditions with reduced evapotranspiration persisted in the north and northeast, while near-normal levels held elsewhere. In March, elevated evapotranspiration is predicted across the country except the central east, where reduced conditions are expected.





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About the India Hydrological Outlook

This document provides a comprehensive water outlook for India, covering the current month, the past four months, and a one-month forecast. The outlook is generated using observational data and an extended-range forecasting system for meteorological variables, combined with advanced hydrological modeling tools. These tools analyze relative changes and predict water availability across various regions, offering valuable insights into hydrological situation in India. Developed by the Water and Climate Lab at IIT Gandhinagar, this outlook aims to support water resource management, planning, and decision-making by offering reliable information on rainfall, temperature, soil moisture (60 cm), runoff, and streamflow patterns across different timescales.

Datasets

The India Hydrological Outlook is made possible through the valuable cooperation of numerous data providers, whose contributions are gratefully acknowledged. Contemporary daily observations and extended range forecast data are supplied by the India Meteorological Department (IMD) https://www.imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html. These datasets are used for initializing hydrological models and generating reliable outlook information through statistical analysis of historical analogues. Hydrological model is calibrated and validated using daily streamflow for sufficient long duration from India-Water Information System https://indiawris.gov.in/wris#RiverMonitoring.

Hydrological Model

The India Hydrological Outlook utilizes the Variable Infiltration Capacity (VIC) model, a large-scale, semi-distributed hydrological model that computes water and energy budgets for each grid. The VIC model captures sub-grid variability in vegetation and elevation, providing a more accurate representation of hydrological processes at a regional scale. VIC model simulations are performed at a daily temporal resolution to provide accurate hydrological forecasts and enhance water resource management across India.

Disclaimer and Liability

The India Hydrological Outlook aims to ensure that all content provided is accurate and aligns with the current scientific understanding. However, the science underlying the meteorological and hydrological forecasts, as well as climate projections, is continuously evolving. As such, any forecast or prediction included in the content should not be regarded as a definitive statement of fact. To the fullest extent permitted by applicable law, the India Hydrological Outlook disclaims all warranties or representations, whether express or implied, regarding the content. Your use of the content is entirely at your own risk, and we make no guarantees that the content is error-free or suitable for your specific needs.

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